

MINI PROJECT LOGBOOK

(CSM501: Mini Project 2 A)

GROUP MEMBERS

Sanchet Khemani (D12B 21)

Atharva Hande (D12B 67)

Aryan Surve (D12A 67)

Tarunkumar Gatla (D12A 71)

Name of the Mentor

Prof. Mrs. Pallavi Saindane



Department of Computer Engineering

Vivekanand Education Society's Institute of Technology,

An Autonomous Institute affiliated to University of Mumbai

HAMC, Collector's Colony, Chembur,

Mumbai-400074

University of Mumbai (AY 2024-25)

INSTITUTE VISION & MISSION

VISION:

To create a vibrant knowledge oriented environment with innovative teaching practices and to inculcate a tradition of socially conscious application of technology.

MISSION:

- To inculcate a culture of value based education.
- To enthuse students to develop in an ambient environment of caring and of sharing information.
- To enable students to work towards excellence in their chosen field with a professional bent of mind.

COMPUTER ENGINEERING DEPARTMENT

VISION:

To reach international standards by empowering students with Computing skills and cutting edge technology

MISSION:

- To sustain excellence in teaching and research and create center of excellence
- To provide broad Educational and Research experiences through interdisciplinary and industrial collaboration programs.
- To prepare students to enter the world of computing and make them ready for productive employment in the public or private sectors, enhance their entrepreneurship skills and motivate them to pursue advanced degrees.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

I	To provide students with a solid foundation in their core concepts of mathematical, scientific and computer engineering fundamentals required to comprehend, analyze and design solutions for real life problems.
II	To inculcate in students, a balanced outlook with professional and ethical attitude, develop effective communication skills, teamwork and leadership qualities with multidisciplinary approach.
III	To prepare students to excel in postgraduate programs through an excellent academic environment and make them ready for productive employment in the public or private sectors and provide lifelong learning experience.
IV	To provide broad educational and research experience through interdisciplinary and industry centric programs.

PROGRAM OUTCOMES (POs)

Program Outcome Code	Program Outcome Description
PO1	Basic Engineering knowledge: An ability to apply the fundamental knowledge in mathematics, science and engineering to solve problems in Computer engineering.
PO2	Problem Analysis: Identify, formulate, research literature and analyze computer engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and computer engineering and sciences
PO3	Design/ Development of Solutions: Design solutions for complex computer engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal and environmental considerations.

PO4	Conduct investigations of complex engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO5	Modern Tool Usage: Create, select and apply appropriate techniques, resources and modern computer engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to computer engineering practice.
PO7	Environment and Sustainability: Understand the impact of professional computer engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of computer engineering practice.
PO9	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
PO11	Project Management and Finance: Demonstrate knowledge and understanding of computer engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	Professional Skills - The ability to develop programs for computer based systems of varying complexity and domains using standard practices.
PSO2	Successful Career - The ability to adopt skills, languages, environment and platforms for creating innovative career paths, being successful entrepreneurs or for pursuing higher studies.

STUDENT INFORMATION

Project Title: Smart Urban Traffic Optimization for Emergency Services and Prioritization System

	Student 1	Student 2	Student 3	Student 4
Roll No.	21	67	67	71
Name	Sanchet Khemani	Atharva Hande	Aryan Surve	Tarunkumar Gatla
Class with Division	D12B	D12B	D12A	D12A
Contact No.	+91 93225 48563	+91 84519 61965	+91 88502 80758	+91 84509 95973
E-mail	2020.sanchet.khemani@ves.ac.in	d2022.atharva.hande@ves.ac.in	d2022.aryan.surve@ves.ac.in	d2022.tarunkumar.gatla@ves.ac.in
Address	1/12 Borla society	B/18 room no. 04	101/903 Labham	Building no.104, 1st floor, room no.1
	Union park, CG	Shivkrupa CHS	Residency, Pantnagar	6th lane , Nagpada
	Road, Chembur,	sec-10 sanpada	Ghatkopar (EAST),	Mumbai - 400008
	Mumbai 400074	Navi Mumbai 400705	Mumbai-400075	

INSTRUCTIONS TO STUDENTS:

1. The logbook must be submitted to the mentor or Co-Mentor for verification and evaluation of project activities at least once in a week.
2. Logbook duly signed by the guide must be submitted with a project report for evaluation at the end of semester to the department.

DECLARATION

I declare that this project represents my ideas in my own words and wherever others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my project work. I promise to maintain minimum 75% attendance, as per the University of Mumbai norms. I understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours Faithfully

1. Sanchet Khemani (D12B 21)
2. Atharva Hande (D12B 67)
3. Aryan Surve (D12A 67)
4. Tarunkumar Gatla (D12A 71)

(Signature of Students)

Letter of Acceptance

I undersigned, **Prof. Mrs. Pallavi Saindane** working in the Computer Engineering department, willing to guide the project titled **Smart Urban Traffic Optimization for Emergency Services and Prioritization System** for the Mini Project 2 A Semester V respectively for the *Academic Year 2024-25*. The names of the students are:

1. **Sanchet Khemani (D12B 21)**
2. **Atharva Hande (D12B 67)**
3. **Aryan Surve (D12A 67)**
4. **Tarunkumar Gatla (D12A 71)**

(Project Guide)

(Mini Project Coordinator)

(HOD Computer)

COURSE OUTCOMES

CO No.	COURSE OUTCOME	POs covered	PSOs covered
CO1	Identify problems based on societal /research needs.	PO1, PO2,PO4	PSO1,PSO2
CO2	Apply Knowledge and skill to solve societal problems in a group.	PO1,PO2,PO4, PO5,PO6,PO8	PSO1,PSO2
CO3	Develop interpersonal skills to work as a member of a group or leader.	PO1,PO2,PO4, PO9,PO11	PSO1,POS2
CO4	Draw the proper inferences from available results through theoretical/ experimental/simulations.	PO1,PO2,PO4, PO5,PO6,PO12	PSO1,POS2
CO5	Analyze the impact of solutions in societal and environmental context for sustainable development.	PO2,PO3,PO4, PO7,PO12	PSO1,POS2
CO6	Use standard norms of engineering practices	PO1,PO2,PO4, PO12	PSO1
CO7	Excel in written and oral communication.	PO1,PO4,PO8, PO9,PO10,PO12	PSO1
CO8	Demonstrate capabilities of self-learning in a group, which leads to lifelong learning.	PO1,PO2,PO4, PO12	PSO1
CO9	Demonstrate project management principles during project work.	PO1,PO2,PO4, PO11,PO12	PSO1,POS2

CO-PO-PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	2	-	-	-	-	-	-	-	-	1	1
CO2	2	2	-	2	3	2	-	2	-	-	-	-	2	1
CO3	1	1	-	2	-	-	-	-	3	3	-	-	1	1
CO4	2	1	-	1	2	2	-	-	-	-	-	2	2	1
CO5	-	2	1	2	-	-	3	-	-	-	-	1	1	2
CO6	1	2	-	1	-	-	-	-	-	-	-	2	2	-
CO7	1	-	-	1	-	-	-	3	2	2	-	1	1	-
CO8	1	3	-	3	-	-	-	-	-	-	-	2	1	-
CO9	1	1	-	2	-	-	-	-	-	-	2	2	1	2

SCHEDULE FOR MINI PROJECT

Date	Week	Contents	Remark	Guide Sign
7/02/25	1	Discuss about Research Paper and Video Processing	Collect Dataset Accordingly	
14/02/25	2	Audio Processing and RFID Technology	Model Creation	
17/02/25	3	Dataset Trained using ROBOFLOW and Model Creation	Mention Algorithms For Priority	
27/02/25	4	Formed and Improvement in Priority based token system	Create an algorithm from the research paper read.	
1/03/25	5	Mini Project Review 1	Integrate traffic signaling with signal automation	
11/03/25	6	Yolo Models Comparison and algorithm studied	Do fine tuning of algo	
17/03/25	7	Performed Multithreading on 4 lanes	Do Hardware Implementation	
21/03/25	8	Hardware implementation and components required	Integrate hardware with software	
26/3/25	9	Integration of Tokens with hardware	Optimize timing of signals	
1/4/25	10	Mini Project Review 2	Search for indian ambulance dataset	
11/4/25	11	Explored various yolo version depending upon their Accuracy on same dataset	Check for their performance	
21/4/25	12	Created and Demonstrated full Traffic Model	Create report	

PROGRESS/ATTENDANCE REPORT

Title of the Project: Smart Urban Traffic Optimization for Emergency Services and Prioritization System	
Group No.43	Sanchet Khemani (D12B 21) Atharva Hande (D12B 67) Aryan Surve (D12A 67) Tarunkumar Gatla (D12A 71)
Name of the Supervisor:	

Sr. No	Date	Attendance				Progress/Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
1	29/9/24	✓	✓	✓	✓	Studied Papers Related Traffic Automation			
2	5/8/24	✓	✓	✓	✓	Studied the papers and Prepared the summary of each paper			
3	19/8/24	✓	✓	✓	✓	Worked on System Design			
4	26/8/24	✓	✓	✓	✓	Implemented our sample on SUMO			
5	2/9/24	✓	✓	✓	✓	Studied Various Algos to Implement Model			
6	16/9/24	✓	✓	✓	✓	Studying various traffic model and algos			
7	23/9/24	✓	✓	✓	✓	Steps of Implementation of Model using ML			
8	10/10/24	✓	✓	✓	✓	Understood various algos for image processing			
9	14/10/24	✓	✓	✓	✓	Explored Video Processing Algorithms			
10	17/10/24	✓	✓	✓	✓	Reviewed Relevant Literature on Video Processing			
11	21/10/24	✓	✓	✓	✓	Analyzed some models and video processing techniques			
12	22/10/24	✓	✓	✓	✓	Implementation of both Models			
13	7/2/25	✓	✓	✓	✓	Multithreading using yolo v8 successful			
14	13/2/25	✓	✓	✓	✓	Fine tuning of Multithreading algorithm			
15	27/2/25	✓	✓	✓	✓	Implemented Priority based token system			
16	1/3/25	✓	✓	✓	✓	Mini Project Review: Integrity of traffic signaling algorithm with traffic module and hardware integration			
17	11/3/25	✓	✓	✓	✓	Captured video of traffic for video processing			
18	21/3/25	✓	✓	✓	✓	IEEE research paper discussed			
19	25/3/25	✓	✓	✓	✓	Hardware implementation and components required			
20	1/4/25	✓	✓	✓	✓	Suggestion: try to optimize efficiency and adoptability			
21	26/4/24	✓	✓	✓	✓	Optimized efficiency and adoptability			

Sign of the Supervisor

EXAMINER'S FEEDBACK FORM

Name of External examiner: _____

College of External examiner: _____

Name of Internal examiner: _____

Date of Examination: ____/____/____

No. of students in project team: 4

Availability of separate lab for the project: Yes / No

Student Performance Analysis (Put Tick as per your Observation)

Excellent (3)		Very Good (2)		Good (1)	
Sr. No.	Observation	(3)	(2)	(1)	
1	Quality of problem and Clarity				
2	Innovativeness in solutions				
3	Cost effectiveness and Societal impact				
4	Full functioning of working model as per stated requirements				
5	Effective use of skill sets				
6	Effective use of standard engineering norms				
7	Contribution of an individual's as member or leader				
8	Clarity in written and oral communication				
9	Overall performance				

- o Can the same mini project extend to next semester by adding new objectives/ideas? (Yes/ No)
- o If yes, suggest new Innovative Technique/Idea/ objectives related to this project.

Signature of External Examiner

Signature of Internal Examiner